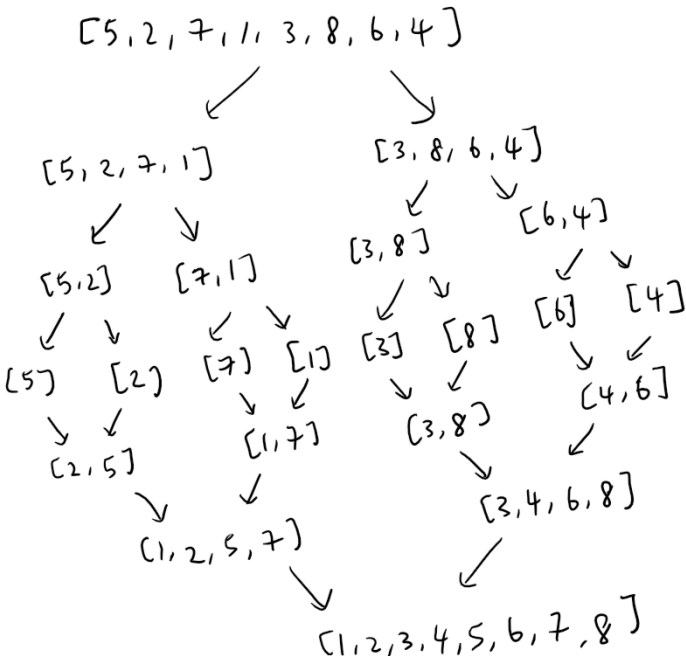
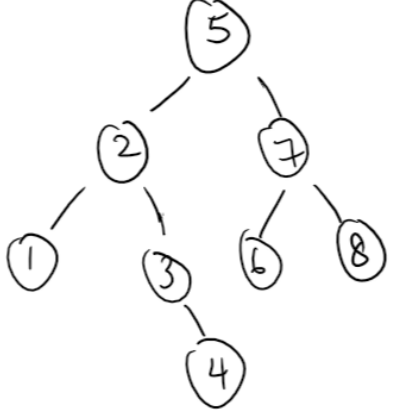


2024 JC1 Promo Exam Paper 1 (Theory) – Suggested Solutions

1(a)	The algorithm starts from the first index, index 0, to find the value there. Since it is not found, it moves on sequentially to the next index, index 1, and index 2, where 42 is found. The search terminates.	1m: start at beginning 1m: search in sequence, found at index 2
(b)	1. The search will at first encompass the entire array from index 0 to 7 2. The middle index is calculated as $(0 + 7) // 2 = 3$, where 51 was found. Since 51 is bigger than the key, the search should continue in the left-half. The search region is now confined from index 0 to 2. 3. The new middle index is calculated as $(0 + 2) // 2 = 1$, where 37 was found. Since 37 is smaller than the key, the search should continue in the right-half. The search region is now confined from index 2 to 2. 4. At index 2, 42 was found, which matches the key. The search stops here.	1m: Initialising and use of indices 1m: Calculating middle index 1m: Eliminating one half
(c)	For this specific case, both algorithms appear to take the same number of steps to find this key 42 due to the specific position of the key in the array. But this does not mean they are equally efficient in general cases with large arrays. Linear search has a time complexity of $O(n)$, eliminating the search space by 1 for each step taken to find the key. Binary search has a time complexity of $O(\lg n)$ which is much faster for large sorted arrays, because it reduces the search space by half with each step/iteration.	1m: Same for particular context 1m: $O(n)$ for linear 1m: $O(\lg n)$ for binary 1m: reduces the search by half

2(a)	The unsorted array is recursively split into two until they are single-element arrays, and then recursively recombining them, sorting the array in the process by comparing the first elements of each sub array. 	1m: split 1m: combining <u>Diagram</u> 1m: show splitting into halves until single element 1m: recombining them into a sorted array
(b)	$O(n \log(n))$	1m
(c)		1m: root node is 5 1m: left subtree correct 1m: right subtree correct
(d)	In-order traversal: Starting from the root, recursively visit the left sub-tree, followed by the node itself, then visit the right subtree.	1m: in-order 1m: left > node > right

3(a)	Seagreen colour: 2E 8B 57 in decimal = $2 \times 16^5 + 14 \times 16^4 + 8 \times 16^3 + 11 \times 16^2 + 5 \times 16^1 + 7 \times 16^0$ $= 2\,097\,152 + 917\,504 + 32\,768 + 2\,816 + 80 + 7$ $= 3\,050\,327$ (Alternative answer accepted): 2E: 8B: 57 = $2 \times 16^1 + 14 \times 16^0 : 8 \times 16^1 + 11 \times 16^0 : 5 \times 16^1 + 7 \times 16^0$ $= 46 : 139 : 87$	1m: Correct base/power 1m: Correct final ans
(b)	Chocolate colour D2:69:1E to binary: 1101 0010 0110 1001 0001 1110	1m: Concept/Calc 1m: Final ans
(c)	- Hexadecimal numbers are of base 16, related to base 2 in digital data. - This aligns well with binary/digital data in computer systems with 4 binary digits forming 1 hexadecimal digit, making it easier to convert to binary values (digital data) directly, Decimal representation requires more complex conversions to digital data. - Hexadecimal can be easier/more intuitive to read than large decimal numbers, improving clarity when examining digital data for debugging. For instance, in Q3a, it is difficult to segment the denary colour (3 050 327) into the separate RGB components as opposed to the hexadecimal form.	1m Alignment to digital data (binary) 1m Intuitive/more useful in digital context

4(a)	A suitable composite PK: {CustomerID, ProductID, Date}	1 mark
4(b)	1. The value of a PK can be used to uniquely identify a particular record in a table. 2. The records in a table are sorted by the PK for easy searching and retrieval.	1 mark each
4(c)(i)	Add a field (or column) in the table and name it OrderID.	1 mark
4(c)(ii)	New PK: {OrderID} Advantage (one of the followings): - Lesser value to provide to specify a record - More efficient in sorting of PK or searching and retrieval of record.	1 mark 1 mark

5(a)	A: Switch B: Router	1 mark each
5(b)	Differences between a switch and a router (any 2 of the followings): 1. Router: • Function: A router connects multiple networks and directs data between them. It determines the best path for data to travel from one network to another. • IP Addressing: Routers use IP addresses to forward data to the correct destination across networks. • Connection: It connects different networks (e.g., local networks to the internet or different subnetworks). 2. Switch: • Function: A switch connects multiple devices within the same network (e.g., computers, printers). It forwards data to the correct device based on its hardware (MAC) address. • MAC Addressing: Switches use MAC addresses to forward data between devices in a local area network (LAN). • Connection: Primarily used to connect devices like PCs, printers, and servers within the same network. Key Differences: • Purpose: Routers connect different networks, while switches connect devices within the same network. • Addressing: Routers use IP addresses, switches use MAC addresses. • Functionality: Routers direct traffic between networks, while switches manage traffic between devices within a single network.	1 mark each
5(c)	One advantage and one disadvantage of using cables to connect networks together: (Any one of the followings) Advantages: 1. Reliability: Wired connections are generally more stable and reliable compared to wireless connections. Cables are less susceptible to interference and signal loss. 2. Speed: Wired connections often offer higher data transfer speeds and lower latency than wireless networks, which can be beneficial for tasks that require high bandwidth.	1 mark each

	3. Security: Physical cables provide a more secure connection because accessing the network requires physical access to the cabling infrastructure. This reduces the risk of unauthorized access compared to wireless networks. 4. Consistency: Wired networks are less affected by environmental factors such as walls, physical obstructions, and electronic interference, leading to more consistent performance. Disadvantages: 1. Cost: Installing and maintaining cables can be expensive, especially if extensive cabling is required or if cables need to be routed through walls and ceilings. 2. Flexibility: Wired connections are less flexible compared to wireless ones. Adding or moving devices requires running additional cables, which can be cumbersome. 3. Maintenance: Physical cables are subject to wear and tear and may require regular maintenance or replacement if damaged. 4. Scalability: Expanding a wired network may require additional cables and network equipment, which can be complex and costly to implement. 5. Aesthetics: Cabling can be unsightly and may require careful planning and installation to maintain a clean and organized appearance, particularly in visible areas.	
5(d)	State and explain one disadvantage of connecting networks together via the internet: ☐ Security Risks and Data Privacy: Connecting networks via the internet can expose them to various security risks, including hacking, data breaches, and cyberattacks.	2 marks
5(e)	Any one of the followings: Implement strong security measures to prevent unauthorised access to the networks, like VPNs (Virtual Private Networks), firewalls, and password protection. Implement data encryption or transmitted data using secured protocol like HTTPS over the internet	1 mark